

22PH202 ELECTROMAGNETISM AND MODERN PHYSICS

UNIT III – LM 3.4 - HUMAN EYE AND ITS DEFECTS

1.1 RANGE OF VISION OF THE HUMAN EYE

Eye is one of our most important sense organs. The eye enables us to see the various objects around us. The main parts of the human eye are : Cornea, Iris, Pupil, Ciliary muscles, Eye lens (which is a flexible convex lens), Retina and Optic nerve. All these parts of the eye are shown in the simplified diagram of human eye given in Figure. We will now describe the construction and working of the human eye.

Human eye is an organ of the human body which is used for vision. It is located in the human skull. Approximate size of a human eye is 2.5 cm in diameter. The important parts of eye are:

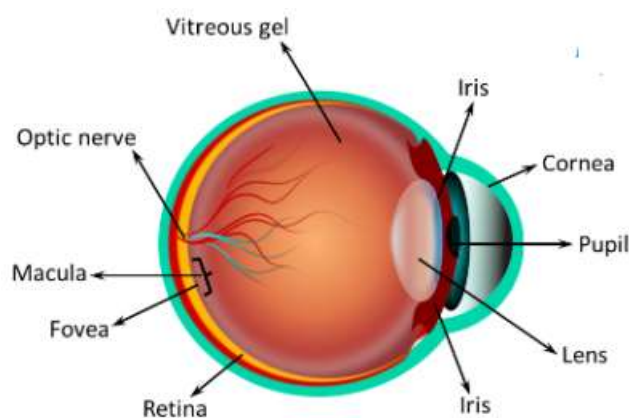


Figure -1 Vision of the human eye

1. Cornea- The transparent portion of the eye that allows the light to enter our eye is known as the 'cornea'. It is made up of transparent tissue. It covers the various parts like the pupil, anterior chamber, and the iris. It refracts light and accounts for two-thirds of the eye's total power. The refractive power of cornea is approximately 43 dioptries.

2. Iris- Iris is made up of contracting and relaxing muscles in the eye. It controls the size of the pupil and the light reaching the retina. Iris defines the color of your eyes.

3. Pupil- Pupil is placed in the center of the iris and allows light to reach the retina. Its size is controlled by the iris. It appears in black color so the eye tissues absorb or diffusely reflect the light entering the pupil.

4. Lens- Biconvex lens of transparent structure present in the eye behind the pupil. The lens refracts the light, so it can focus on the retina. The lens is capable of changing the focal distance by changing its size. When Eye sees the object at infinity it is said to be in its relax mode.

5. Retina- After refracting the light from the lens it comes on a light-sensitive tissue called retina, which is in the inner coat of the eye that sends electrical signals after converting them from light to the brain for processing. Two light-sensitive cells are present in the retina - one is Rods and other cones. Rods for night-time vision and cones to see the colors.

6. Optic Nerves- The optic nerve is used to send electrical impulses from the retina to the brain.

1.2 FAR POINT AND NEAR POINT

Distance of the farthest object that can be seen clearly by naked eye is called the Far point of eye. For the normal human eye it is at infinity.

Closest distance of an object at which it can be seen clearly by the naked eye is known as the Near point of eye. It is denoted by 'D' and for a normal human eye it is 25 cm.

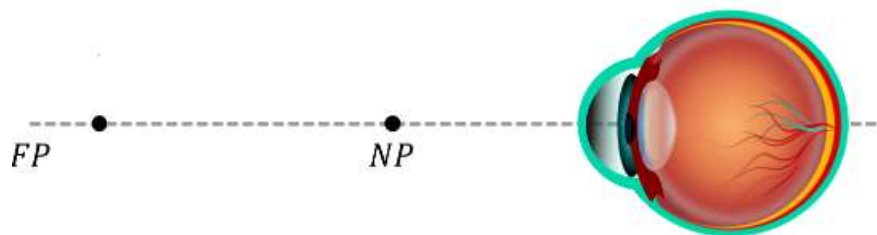
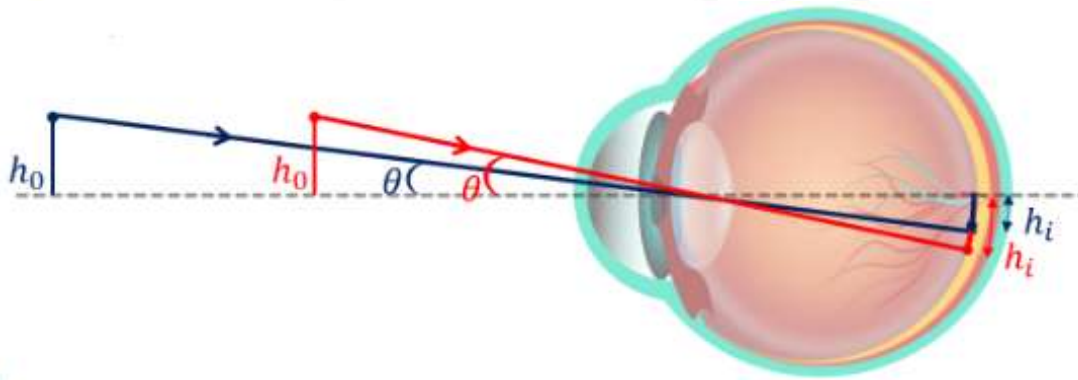


Figure -2 Far Point And Near Point Of The Human Eye

1.2.1 Apparent size

The size of an object as sensed by us depends on the size of the image formed on the retina. As shown in figure the object which is near to the eye forms a bigger size of image than the object which is away from the eye.



1.3 DEFECTS OF HUMAN EYE AND CORRECTION

Defect in the eye means that the proper image is not formed on the retina. Here are some common defects of eye:

1.3.1 EMMETROPIA

The following happens in emmetropic eyes:

1. Light enters the eyes via the pupil (the black hole at the center) and focuses on the **retina**. The **retina** is the light-sensitive part at the back of the eye. It contains special cells known as photoreceptors (rods and cones).³
2. These photoreceptors receive and transform light into electrical signals, which are then transmitted to the brain via the optic nerve.
3. In the brain, the signals are interpreted, enabling you to see clearly and understand what you see.

If a person has emmetropic eyes, light rays entering the eye from a distance are directly focused on the retina, resulting in normal vision (emmetropia), otherwise, they will have ametropia.

1.3.2 Myopia or Near-sightedness- It is a common vision condition in which you can see objects nearby clearly, but objects farther away are blurry. It occurs when the shape of the human eye causes light rays to bend (refract) incorrectly, focusing images in front of the retina instead of on the retina. The far point for this eye is closer than infinity.

For the correction of this defect we wear contact glasses made of concave lenses.

It diversifies the light rays so they can focus on the retina.

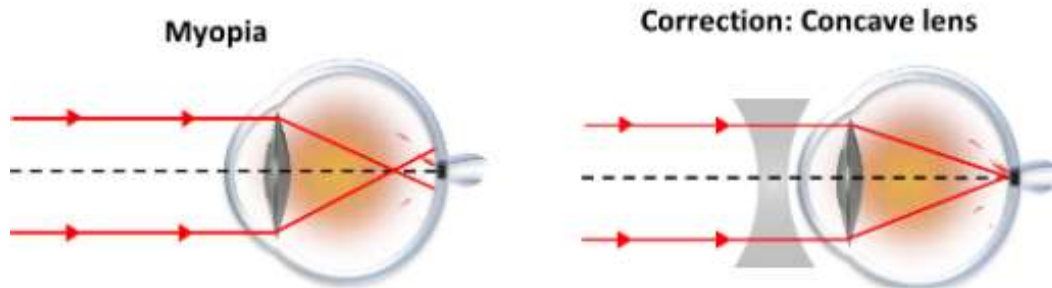


Figure -3 Myopia or Near-sightedness Of The Human Eye

Myopia is commonly genetic, meaning it's passed down the family line. It usually affects children between ages 6 and 14 and worsens as they age.⁵

Other symptoms include:

- Eye strain leading to headaches
- Poor night vision
- Needing to squint to see clearly (e.g., when watching television)
- Excessive blinking
- Eye rubbing

1.3.3. HYPERMETROPIA OR FAR-SIGHTEDNESS-

It is a condition of the eye where distant objects are seen clearly but nearby objects appear blurred. It is due to incoming light being focused behind, instead of on the retina. The near point for this eye increases.

Hypermetropic eye

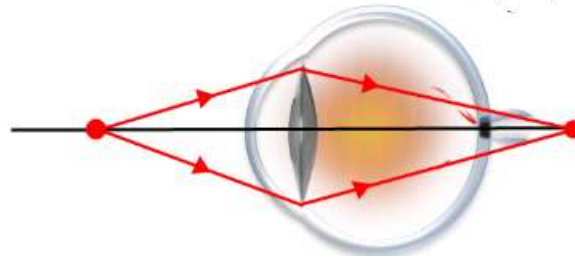
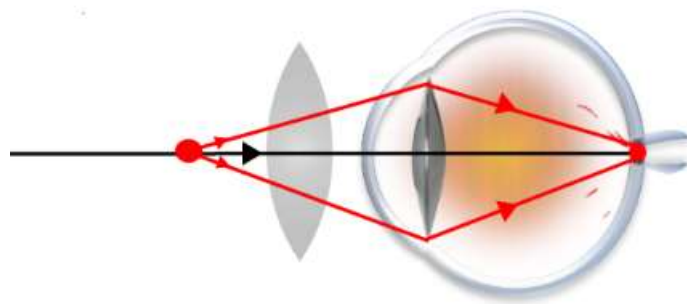


Figure -4 Hypermetropia Of The Human Eye

The correction of hypermetropia can be done by wearing glasses made of convex lens. It will converge the light ray and focus on the retina.

Convex lens



Hyperopia results from an abnormal corneal curvature or shorter eyeball (from front to back). This condition is genetic and usually presents at birth. Other symptoms include:

- Needing to squint to see clearly
- Eye strain
- Burning/itching/aching eyes
- Headaches after close tasks such as reading or computer work

1.3.4 . ASTIGMATISM-

In Astigmatism the light rays entering the eye do not focus evenly to a single focal point on the retina but instead are scattered. Some light rays focus on the retina and some focus in front of or behind it. This is because of non-uniform curvature of the cornea; resulting in a distorted or blurry vision at any distance.

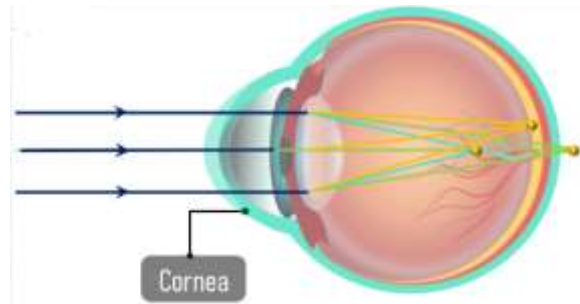


Figure – 5 Astigmatism Of The Human Eye

Someone with mild astigmatism may not notice any symptoms. This is why regular eye check-ups are important, especially for children who are less likely to realize visual abnormalities. Common symptoms include:

- Blurry vision
- The need to squint to see clearly
- Eye strain accompanied by headaches
- Poor night vision

1.4 HOW TO PROTECT YOUR EYES

Whether you have emmetropia or ametropia, protecting your eyes is important. You can protect your eyes and maintain healthy vision by:

- Attending regular eye check-ups, even if you have good eyesight (mild symptoms may go unnoticed)
- Managing chronic conditions such as diabetes and hypertension, which can affect vision
- Wearing hats and UV sunglasses to prevent direct sunlight
- Using protective gear when playing a contact sport or engaging in risky activities such as mowing the lawn
- Avoiding smoking

- Using corrective lenses to optimize your vision
- Avoiding **digital eye strain** by observing the **20-20-20 rule**
- Avoiding poor lighting, which can strain your eyes

Visit your ophthalmologist, optometrist, or optician urgently if you experience blurry vision, double vision, flashes of light, or halos around lights. These may indicate a serious medical or eye condition.

Problems: 1

A person cannot see objects beyond 80 cm, whereas a normal person can easily see an object distant 160 cm. What is the focal length and nature of the lens used to rectify this defect?

Answer: Given $u = -160$

$$v = -80$$

Using lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{-80} - \frac{1}{-160} = \frac{1}{f}$$

$$\frac{1}{160} - \frac{1}{80} = \frac{1}{f}$$

$$\frac{1}{f} = \frac{1-2}{160}$$

$$f = -160 \text{ cm}$$

Question 1. What is the cause for Myopia or short- sightedness?

Answer: It happens due to the eyeball being too long and the high converging power of the eye.

Question 2. What is the advantage of having two eyes?

Answer: When one eye is open, a human being can see a horizontal field of view of about

150° but with two eyes open, the field of view is 180°. Thus with two eyes we can see a wider field of view.

Question 3. What is the nature of the image formed on the retina?

Answer: Inverted image is formed on the retina, but our brain helps in interpreting it as an erect image.

Question 3. What is night blindness?

Answer: Night Blindness is a retina problem which prevents you from seeing well at night or in poor lighting.